



Two networks. One control plane.

Fixed access and mobile access are managed by two disjoint toolchains, two data models and two on-call rotations. Aether puts them behind one.

10 PROTOCOLS · ONE BINARY · OPEN AGENT · MANAGED PLATFORM

THE PROBLEM

The access network is managed twice.

Every operator runs both of these. Almost nobody runs them from one place.

CPE / WiFi

- uCentral · TIP OpenWiFi
- TR-369 / USP
- WRP · RDK-B
- IEEE 1905.1 EasyMesh
- OpenSync

BROADBAND TEAM

Transport / RAN

- NETCONF + YANG
- SNMP v1/v2c/v3
- gNMI streaming
- O-RAN A1 policy
- VES event collector

RAN + TRANSPORT TEAM

When a subscriber says “my connection is bad” —

no single system can see the home, the backhaul and the radio.

PROTOCOL COVERAGE

Ten protocols. One binary.

uCentral / TIP	WebSocket + JSON · OpenWiFi, OpenWrt
TR-369 / USP	WebSocket + MQTT · protobuf · QSDK, prpIOS
WRP / Xmidt	RDk-B, zero-install via Parodus
IEEE 1905.1	EtherType 0x893a · CMDU · EasyMesh R1–R5
OpenSync	MQTT · Plume migration path
MQTT	3.1 / 5.0 clustered broker · USP MTP
NETCONF / YANG	SSH + XML · RFC 6241 / 6242
SNMP	v1 / v2c / v3 · polling + traps
gNMI	gRPC + protobuf · OpenConfig telemetry
O-RAN A1 / VES	A1 policy → Near-RT RIC · VES collector

ABOVE THE LINE: CPE. BELOW: TRANSPORT, BACKHAUL AND RAN.

PRECISION

Aether is not a RIC. Here is what it is.

IS

- A1 client — deliver, withdraw, fetch and status policy
- VES collector — events into ONAP-style OSS
- Device.Cellular — IMEI, IMSI, RSRP, RSRQ, SINR on FWA CPE
- One data model across CPE, transport and RAN

ON THE ROADMAP

- O1 interface — the NETCONF/YANG transport is already in place
- Non-RT RIC / rApp runtime over R1
- STOMP and CoAP message transports

IS NOT

- A Near-RT RIC, or E2 termination
- An xApp hosting platform
- A replacement for your RAN vendor's EMS

The hard part was never A1. It was a data model where a TR-181 parameter from an OpenWrt box and a gNMI subscription from an aggregation switch are both first-class.

OPEN SOURCE

ac-client.

Download. Install. Sign up.

The agent is open source under BSD 3-Clause. The platform it reports to is a managed service. Three commands and the router is under management.

```
opkg install ac-client luci-app-aclient
uci set optimacs.@config[0].claim_token='...'
/etc/init.d/ac-client start
```

TR-369 / USP 1.3	TP-469 conformance tested
Post-quantum mTLS	X25519 + ML-KEM-768, always on
UCI-backed TR-181	47 backend operations
WiFi 7 + Cellular	EHT modes and Device.Cellular

1,500+ OpenWrt router models. No custom firmware image.

DEPLOYMENT

Meet the device where it is.

No reflash. No certification cycle. No truck roll.

- **RDK-B**
Terminates the Parodus/WRP session already running.
No firmware change.
- **OpenWrt**
opkg package install. 30k+ router models.
- **QSDK**
Agent wrapping cfg80211tool and qca-hostapd.
- **prpIOS**
Speaks the TR-181 data model bus already present.
- **OpenWiFi**
Native uCentral client.
- **Aggregation gear**
NETCONF or gNMI. SNMP for anything older.

Config changes ship as JSON Patch (RFC 6902)

~200 bytes per update, not a ~50 KB full-config push.



Ten protocols. One control plane.

Built for a messy access network. Multiple CPE silicon vendors, a platform inherited from an acquisition, aggregation gear still on SNMP, and fixed wireless CPE nobody has a dashboard for.

- \$0.30 / device / month at volume — no per-device AI tax
- Open source agent, managed platform, EU / US data residency
- Self-hosting licensed on Enterprise

aether-io.com

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